



Trinity Grammar School

Mathematics Department

2015

HALF-YEARLY EXAMINATION

HSC ASSESSMENT TASK 3

Year 12

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- BOSTES approved calculators for this course may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11–16, show relevant mathematical reasoning and/or calculations
- Write your BOSTES Student Number on this question paper and on any answer sheets or writing booklets used to write your responses
- If you do not attempt a question you must submit an answer sheet or writing booklet for that question clearly indicating N/A and your BOSTES Student Number
- HSC Assessment Weighting: 30%

BOSTES Student Number

--	--	--	--	--	--	--	--

Class Teacher:

.....

Do NOT write solutions to be marked on this question paper. Any working on this question paper will NOT be marked.

Date of Examination:
Wednesday 22 April 2015

Total marks – 100

Section I

10 marks

- Attempt Questions 1–10
- Allow about **15** minutes for this section

Section II

90 marks

- Attempt Questions 11–16
- Allow about **2 hours 45** minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Shade the correct response on the multiple-choice answer sheet for Questions 1–10

Each question is worth 1 mark

- 1 If $\sin x = e^y$ for $0 < x < \pi$, what is $\frac{dy}{dx}$ in terms of x ?
- (A) $-\tan x$
- (B) $-\cot x$
- (C) $\cot x$
- (D) $\tan x$
- 2 If n is a non-negative integer, then $\int_0^1 x^n dx = \int_0^1 (1-x)^n dx$ for
- (A) n even, only
- (B) n odd, only
- (C) non-zero n , only
- (D) all n
- 3 If $\frac{d}{dx} f(x) = g(x)$ and $\frac{d}{dx} g(x) = f(x^2)$, then $\frac{d^2}{dx^2} f(x^3) =$
- (A) $6x^2 f(x^6) + x^3 g(x^6)$
- (B) $9x^4 g(x^3) + 3x^2 f(x^3)$
- (C) $9x^4 f(x^3) + 3x^2 g(x^3)$
- (D) $9x^4 f(x^6) + 6xg(x^3)$

4 If $\int x^2 \cos x \, dx = f(x) - \int 2x \sin x \, dx$, then $f(x) =$

(A) $2 \sin x + 2x \cos x + c$

(B) $x^2 \sin x + c$

(C) $2x \cos x - x^2 \sin x + c$

(D) $4 \cos x - 2x \sin x + c$

5 Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos^2(2x)}{4x^2}$.

(A) 0

(B) 1

(C) 2

(D) 4

6 $\int_1^3 \frac{x^2 - 2x + 4}{x + 1} dx =$

(A) $-2 + 7 \ln 2$

(B) $\ln 2$

(C) $7 \ln 2$

(D) $2 + 7 \ln 2$

Question 7 commences on the next page

7 Find $\int \frac{dx}{(x-1)(x+3)}$.

(A) $\frac{1}{4} \ln \left| \frac{x-1}{x+3} \right| + c$

(B) $\frac{1}{4} \ln \left| \frac{x+3}{x-1} \right| + c$

(C) $\frac{1}{2} \ln |(x-1)(x+3)| + c$

(D) $\ln |(x-1)(x+3)| + c$

8 Which pair of equations give the directrices of the hyperbola defined parametrically by $x = 5 \sec \theta$, $y = 2 \tan \theta$?

(A) $x = \pm \frac{1}{\sqrt{29}}$

(B) $x = \pm \frac{25}{\sqrt{29}}$

(C) $y = \pm \frac{2}{5}x$

(D) $y = \pm \frac{5}{2}x$

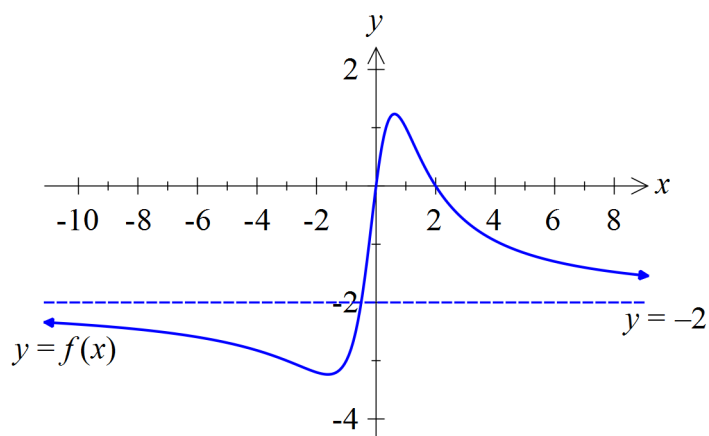
9 Let z represent a complex number with $\arg z = -\frac{\pi}{3}$ and $|z| = 2$. The argument **and** modulus of the complex number $\frac{2i}{z}$ **respectively** are

(A) $-\frac{5\pi}{6}, \frac{1}{2}$

(B) $-\frac{\pi}{6}, \frac{1}{2}$

(C) $\frac{\pi}{6}, 1$

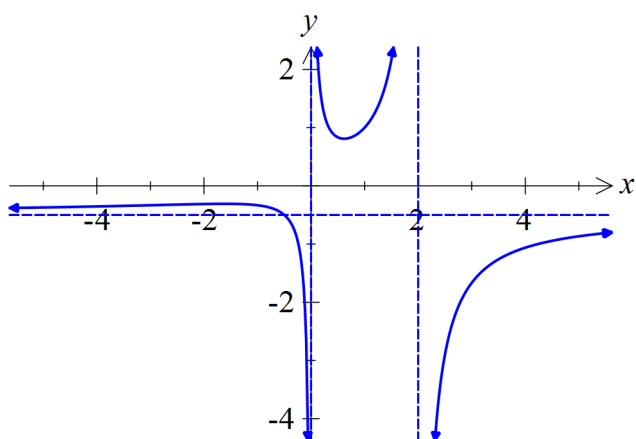
(D) $\frac{5\pi}{6}, 1$



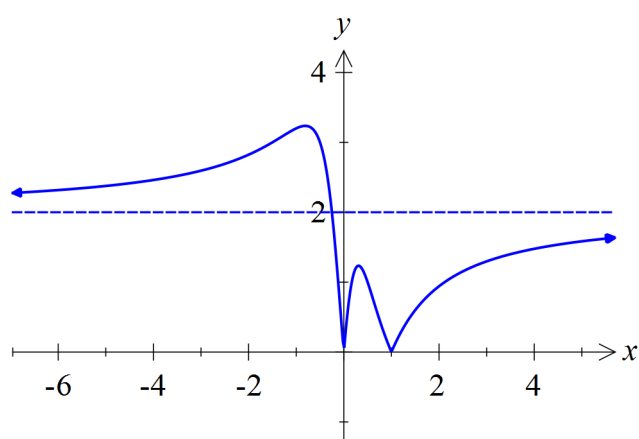
The diagram above represents the graph of $y = f(x)$.

Which diagram below best represents the graph of $y = \frac{1}{f(2x)}$?

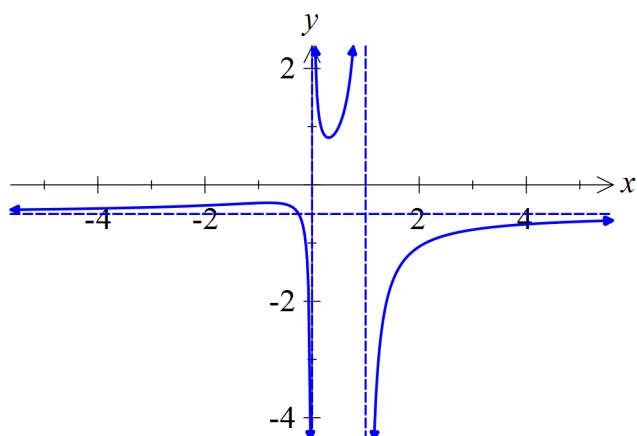
(A)



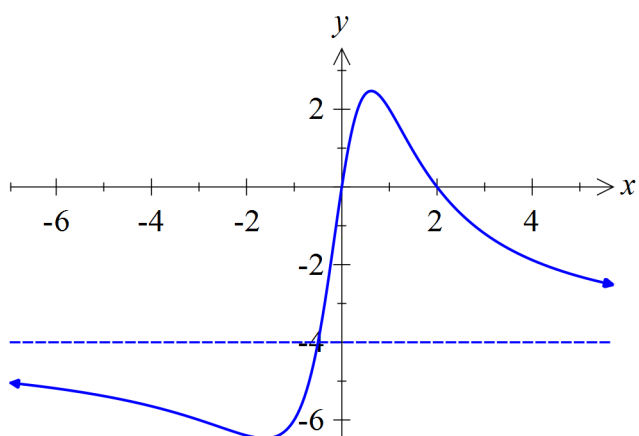
(B)



(C)



(D)



End of Section I
Section II commences on the next page

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include all relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet

(a) Let $z = 5 + 5i$ and $w = 2 + i$.

Find:

(i) $2z + 3\bar{w}$; 2

(ii) $z(w - 1)$; 2

(iii) $\frac{z}{w}$, 2

(iv) $\arg z - \arg \bar{z}$. 1

(b) Using De Moivre's theorem, simplify the expression $\frac{(1 + \sqrt{3}i)^6}{(1 - i)^4}$. 3

(c) If ω is the complex cube root of unity, evaluate $(1 - 3\omega + \omega^2)(1 + \omega - 8\omega^2)$. 2

(d) Sketch in the Argand plane the region satisfying the following pair of inequalities simultaneously. 3

$$\begin{cases} |z - 1 - 2i| \leq 2 \\ \frac{\pi}{4} < \arg z \leq \frac{\pi}{2} \end{cases}$$

End of Question 11

Please turn over

Question 12 (15 marks) Use a SEPARATE writing booklet

- (a) Using De Moivre's theorem or otherwise, find all six roots of the equation $z^6 = -64$. **3**

- (b) Evaluate $\int_0^{\frac{\pi}{2}} \sin^3 x \, dx$. **3**

- (c) (i) Find real constants A , B and C such that **3**

$$\frac{x^2 + x + 1}{x^3 + 3x^2} \equiv \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x + 3}.$$

- (ii) Hence, find $\int \frac{x^2 + x + 1}{x^3 + 3x^2} dx$. **2**

- (d) (i) Express $x^2 + x + 1$ in the form $(x + b)^2 + c$, where b and c are real numbers. **1**

- (ii) Hence, by using your result in (i) and the table of standard integrals or otherwise, find: **1**

$$\int \frac{1}{x^2 + x + 1} dx$$

- (iii) Find $\int \frac{dx}{2 + \sin x}$ using the substitution $t = \tan \frac{x}{2}$. **2**

End of Question 12

Please turn over

Question 13 (15 marks) Use a SEPARATE writing booklet

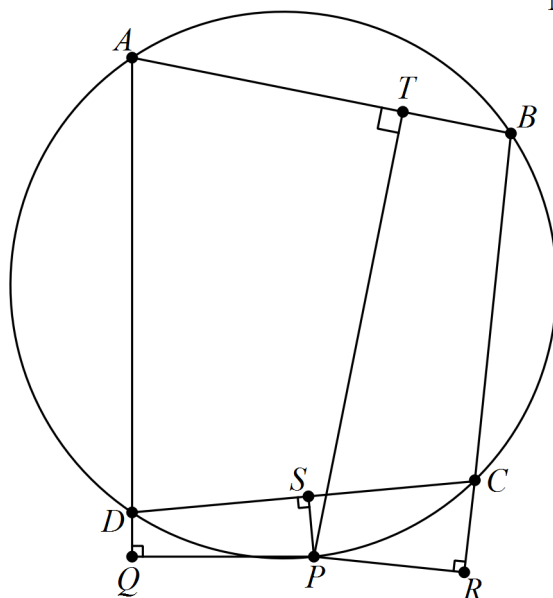
- (a) The variable point $P\left(ct, \frac{c}{t}\right)$ lies on the hyperbola $xy = c^2$.
- (i) Prove that the equation of the tangent to the hyperbola at P is given by: 2
 $x + yt^2 = 2ct$

The perpendicular (from the origin O) meets the tangent at P , at the point N .

- (ii) Prove that the coordinates of N are given by $\left(\frac{2ct}{1+t^4}, \frac{2ct^3}{1+t^4}\right)$. 2
- (iii) Prove that the locus of N as P moves on the hyperbola is the curve defined by $(x^2 + y^2)^2 = 4c^2xy$. 3

(b)

NOT TO SCALE



In the diagram, $ABCD$ is a cyclic quadrilateral. P is a point on the circle through A , B , C and D . PQ , PR , PS and PT are the perpendiculars from P to AD produced, BC produced, CD and AB respectively. **Copy this diagram into your writing booklet.**

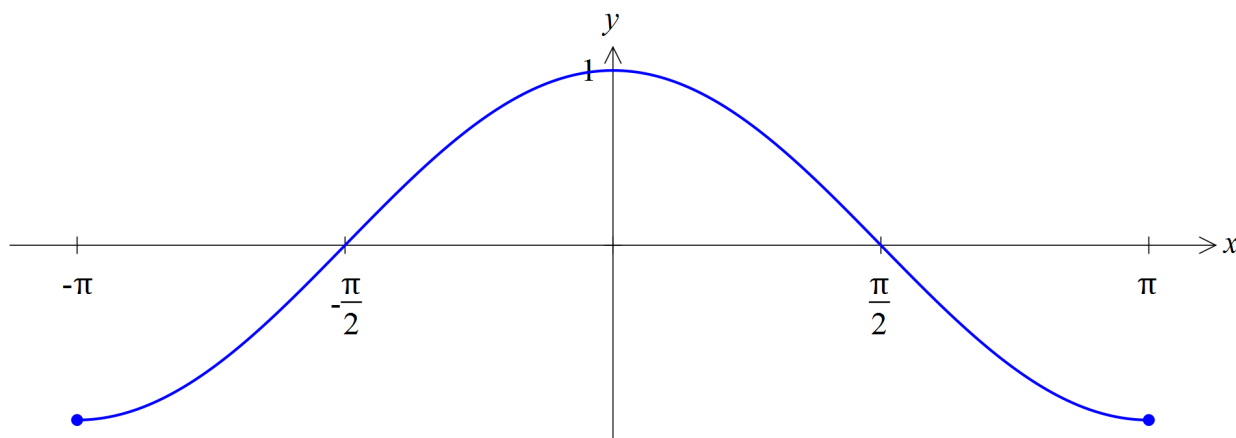
- (i) Explain why $SPRC$ and $AQPT$ are cyclic quadrilaterals. 2
- (ii) Hence show that $\angle SPR = \angle QPT$ and $\angle PRS = \angle PTQ$. 2
- (iii) Prove that $\triangle SPR \parallel \triangle QPT$. 2
- (iv) Hence show that:
- (α) $PS \times PT = PQ \times PR$; 1
- (β) $\frac{PS \times PR}{PQ \times PT} = \frac{SR^2}{QT^2}$. 1

End of Question 13

Please turn over

Question 14 (15 marks) Use a SEPARATE writing booklet

- (a) The diagram below represents the graph (not to scale) of $f(x) = \cos x$ in the interval $-\pi \leq x \leq \pi$.



On separate, one-third page number planes, sketch the graphs of the following:

- | | |
|--|---|
| (i) $y = f(x)^2$ | 2 |
| (ii) $y = \frac{1}{f\left(x - \frac{\pi}{2}\right)}$ | 2 |
| (iii) $y^3 = f(x)$ | 2 |
| (iv) $y = \ln f(x)$ | 2 |
| (v) $y = f(\sqrt{ x })$ | 2 |
- (b) A locus definition of an ellipse is given by $PS + PS' = 2a$, where P is a variable point (x, y) on the ellipse and S and S' represent the foci of the ellipse and $|a|$ is the length of its semi-major axis. Suppose P is represented by the complex variable $z = x + yi$, where x and y are real numbers.

Using the above statement or otherwise, find the eccentricity **and** the Cartesian equation of the ellipse determined by

$$|z - 1| + |z + 1| = 6.$$

(Leave your answer in the Standard form)

End of Question 14

Please turn over

Question 15 (15 marks) Use a SEPARATE writing booklet

- (a) If z is a complex number so that $|z| = 3$ and $\arg z = \frac{\pi}{6}$, **mark clearly on the same Argand diagram** the **points** represented by the following complex numbers:

- (i) z ; 1
- (ii) $-i^{25}z$; 1
- (iii) $z\bar{z}$; 1
- (iv) $z^2 + z$; and 1
- (v) $z^2 - z$. 1

- (b) Sketch the locus in the Argand plane of the variable point z , where $z = x + yi$ such that $|z - 1| = \operatorname{Re}(z)$ **and state** the locus of z in Cartesian form. 2

- (c) For the hyperbola H: $\frac{y^2}{4} - \frac{x^2}{12} = 1$,

- (α) Find:
- (i) its eccentricity; 1
 - (ii) the coordinates of its foci; 1
 - (iii) the equations of its directrices; 1
 - (iv) the equations of its asymptotes. 1
- (β) Sketch the graph of H, showing the features above including its vertices. 2

- (d) Find $\int_1^2 \frac{e^{\frac{1}{x}}}{x^2} dx$. 2

End of Question 15

Please turn over

Question 16 (15 marks) Use a SEPARATE writing booklet

- (a) The points P and Q with respective parameters θ and $\theta + \frac{\pi}{2}$ both lie on the ellipse

$$E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

- (i) Show that Q has coordinates $(-a \sin \theta, b \cos \theta)$. 2
- (ii) If O is the centre of the ellipse E prove that $OP^2 + OQ^2 = a^2 + b^2$. 1
- (iii) Show that the midpoint M of PQ lies on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{1}{2}$. 1
- (iv) State the equation, in parametric form, of the tangent at P on E . 1
- (v) Determine the point of intersection T of the tangent at P and Q to E . 2
- (vi) Show that T lies on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 2$. 1

- (b) The sum of the first k positive integers can be written as 3

$$1 + 2 + 3 + \dots + k = \frac{k(k+1)}{2}.$$

Given that n is a positive integer **prove** that the sum of the integers between 1 and $15n$ inclusive **which are not** divisible by 3 or 5 is $60n^2$.

- (c) (i) If $\sin(x - \theta) = k \sin(x + \theta)$, show that $\tan x = \frac{-(k+1)}{k-1} \tan \theta$. 2
- (ii) Hence, for $k = \frac{1}{2}$ and $\theta = 210^\circ$ solve for x in the interval $0^\circ \leq x \leq 360^\circ$. 2

End of Question 16
End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

Note $\ln x = \log_e x, \quad x > 0$



Trinity Grammar School
Mathematics Department

2015

Year 12 Mathematics Extension 2

HSC Assessment Task 3

Section I

Objective Response Answer Sheet

*Suggested
solutions &
making sc sense*

BOSTES Student Number

--	--	--	--	--	--	--	--	--

Class Teacher: _____

EH

- Write your answers for **Section I** on this answer sheet.
- **Shade completely only ONE answer for each question.**
- To change an answer, erase your previous mark completely and then record your new answer.

1	(A)	(B)	☒	(D)
2	(A)	(B)	(C)	☒
3	(A)	(B)	(C)	☒
4	(A)	☒	(C)	(D)
5	(A)	☒	(C)	(D)
6	☒	(B)	(C)	(D)
7	☒	(B)	(C)	(D)
8	(A)	☒	(C)	(D)
9	(A)	(B)	(C)	☒
10	(A)	(B)	☒	(D)

A table of Standard Integrals is printed on the reverse side of this answer sheet

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Shade the correct response on the multiple-choice answer sheet for Questions 1–10

Each question is worth 1 mark

- 1 If $\sin x = e^y$ for $0 < x < \pi$, what is $\frac{dy}{dx}$ in terms of x ?

(A) $-\tan x$

(B) $-\cot x$

(C) $\cot x$

(D) $\tan x$

$$y = \ln(\sin x)$$

$$\therefore y' = \frac{\cos x}{\sin x} = \cot x$$

- 2 If n is a non-negative integer, then $\int_0^1 x^n dx = \int_0^1 (1-x)^n dx$ for

(A) n even, only

(B) n odd, only

(C) non-zero n , only

(D) all n

$$RHS = \int_0^1 (1-x)^n dx$$

$$\text{let } u = 1-x \quad \therefore du = -dx$$

when $x=0, u=1$
 $x=1, u=0$

$$\therefore \int_0^1 (1-x)^n dx = \int_1^0 u^n - du$$

$$= \int_0^1 u^n du$$

$$= \int_0^1 x^n dx$$

$$\textcircled{02} RHS = \int_0^1 (1-x)^n dx$$

$$= \frac{(1-x)^{n+1}}{-(n+1)} \Big|_0^1 = \frac{1}{n+1}$$

- 3 If $\frac{d}{dx} f(x) = g(x)$ and $\frac{d}{dx} g(x) = f(x^2)$, then $\frac{d^2}{dx^2} f(x^3) =$

(A) $6x^2 f(x^6) + x^3 g(x^6)$

(B) $9x^4 f(x^3) + 3x^2 g(x^3)$

(C) $9x^4 f(x^3) + 3x^2 g(x^3)$

(D) $9x^4 f(x^6) + 6xg(x^3)$

$$\text{Given } f'(x) = g(x) \quad \therefore f'(x^3) = g(x^3)$$

$$g'(x) = f(x^2) \quad \therefore f''(x^3) = g'(x^3)$$

$$\text{let } y = f(x^3)$$

$$\therefore y' = 3x^2 f'(x^3)$$

$$y'' = f'(x^3) \cdot 6x + 3x^2 f''(x^3) \cdot 3x^2$$

$$= 9x^4 f''(x^3) + 6x f'(x^3)$$

$$= 9x^4 f(x^6) + 6xg(x^3)$$

4 If $\int x^2 \cos x \, dx = f(x) - \int 2x \sin x \, dx$, then $f(x) =$

(A) $2 \sin x + 2x \cos x + c$

☒ (B) $x^2 \sin x + c$

(C) $2x \cos x - x^2 \sin x + c$

(D) $4 \cos x - 2x \sin x + c$

let $u = x^2 \quad dv = \cos x$
 $du = 2x \quad v = \sin x$

$\therefore I = uv - \int v \, du$
 $= \underbrace{x^2 \sin x}_{f(x)} - \int 2x \sin x \, dx$

5 Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos^2(2x)}{4x^2}$

(A) 0

☒ (B) 1

(C) 2

(D) 4

$= \lim_{x \rightarrow 0} \left[\frac{\sin^2(2x)}{4x^2} \right]$
 $= \lim_{x \rightarrow 0} \left(\frac{\sin 2x}{2x} \right) \cdot \lim_{x \rightarrow 0} \left(\frac{\sin 2x}{2x} \right)$
 $= 1 \times 1$
 $= 1$

6 $\int_1^3 \frac{x^2 - 2x + 4}{x+1} \, dx =$ $\int_1^3 (x-3) + \frac{7}{x+1} \, dx$ (long division or by inspection)

☒ (A) $-2 + 7 \ln 2$

(B) $\ln 2$

(C) $7 \ln 2$

(D) $2 + 7 \ln 2$

$= \frac{x^2}{2} - 3x + 7 \ln(x+1) \Big|_1^3$
 $= \frac{9}{2} - 9 + 7 \ln 4 - \frac{1}{2} + 3 - 7 \ln(2)$
 $= -2 + 7 \ln 2$

Question 7 commences on the next page

7 Find $\int \frac{dx}{(x-1)(x+3)}$ = $\int \frac{\frac{1}{4}}{x-1} - \frac{\frac{1}{4}}{x+3} dx$

(A) $\frac{1}{4} \ln \left| \frac{x-1}{x+3} \right| + c$ = $\frac{1}{4} [\ln |x-1| - \ln |x+3|]$

(B) $\frac{1}{4} \ln \left| \frac{x+3}{x-1} \right| + c$

(C) $\frac{1}{2} \ln |(x-1)(x+3)| + c$

(D) $\ln |(x-1)(x+3)| + c$

NB: $\frac{A}{x-1} + \frac{B}{x+3} = \frac{1}{(x-1)(x+3)}$

$A(x+3) + B(x-1) = 1$

Let $x=1 \rightarrow A = \frac{1}{4}$
 $x=-3 \rightarrow B = -\frac{1}{4}$ } etc.

8 Which pair of equations give the directrices of the hyperbola defined parametrically by $x = 5 \sec \theta$, $y = 2 \tan \theta$? (a > b)

(A) $x = \pm \frac{1}{\sqrt{29}}$

(B) $x = \pm \frac{25}{\sqrt{29}}$

(C) $y = \pm \frac{2}{5}x$

(D) $y = \pm \frac{5}{2}x$

$\frac{x^2}{25} - \frac{y^2}{4} = 1$

$x = \pm \frac{a}{e} = \pm \frac{5}{e}$

$b^2 = a^2(e^2 - 1)$ $e > 1$

$4 = 25(e^2 - 1)$

$\frac{4}{25} + 1 = e^2$

$e^2 = \frac{29}{25} \therefore e = \frac{\sqrt{29}}{5}$

$\therefore x = \pm \frac{5}{\frac{\sqrt{29}}{5}} = \pm \frac{25}{\sqrt{29}}$

9 Let z represent a complex number with $\arg z = -\frac{\pi}{3}$ and $|z| = 2$. The argument and

modulus of the complex number $\frac{2i}{z}$ respectively are

(A) $-\frac{5\pi}{6}, \frac{1}{2}$

(B) $-\frac{\pi}{6}, \frac{1}{2}$

(C) $\frac{\pi}{6}, 1$

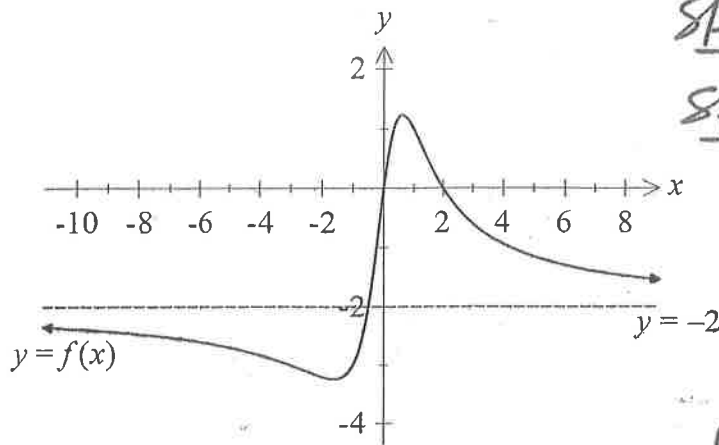
(D) $\frac{5\pi}{6}, 1$

$z = 2 \cos(-\pi/3)$

$\Rightarrow \frac{1}{z} = \frac{1}{2} \cos(\pi/3)$

$\Rightarrow \frac{i}{z} = \frac{1}{2} \cos(\pi/3 + \pi/2)$
 $= \frac{1}{2} \cos(\frac{5\pi}{6})$

$\therefore \frac{2i}{z} = \cos(\frac{5\pi}{6})$



Step 1: Draw $y = f(2x)$

Step 2: Draw $y = \frac{1}{f(2x)}$

In Step 1:

* Same shape but
x-coordinates are halved
at intercepts
y-int is maintained.

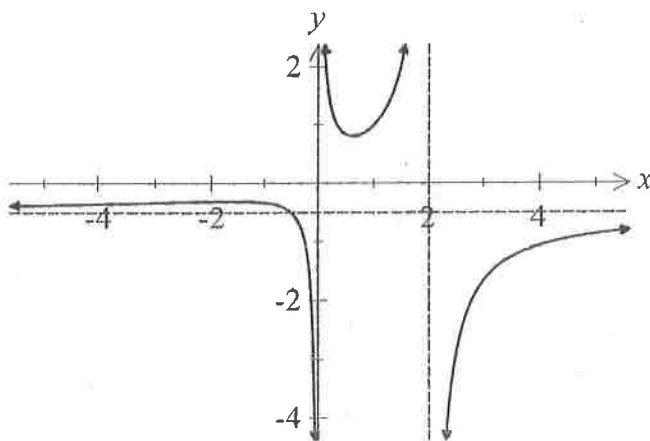
In Step 2: V. Asymptotes
at $x=1$ and $x=0$.

$\therefore$ C

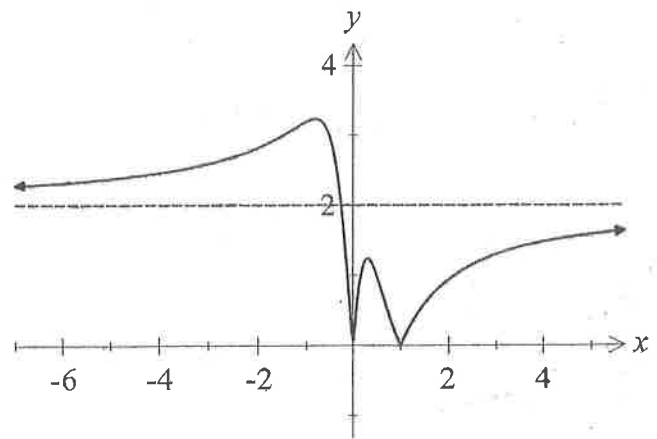
The diagram above represents the graph of $y = f(x)$.

Which diagram below best represents the graph of $y = \frac{1}{f(2x)}$?

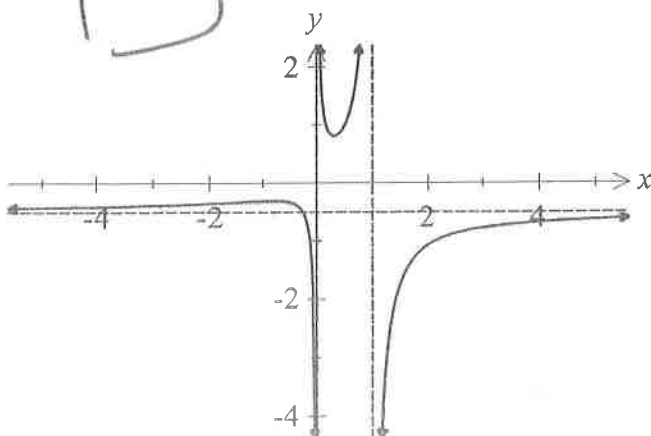
(A)



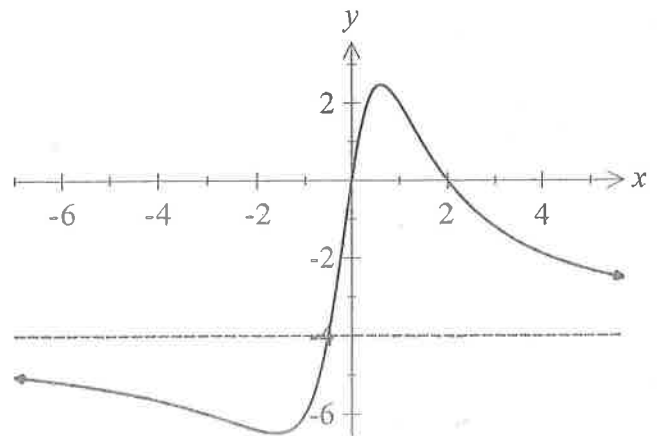
(B)



(C)



(D)



End of Section I

Section II commences on the next page

2015 MAXX Half-Yearly Task 3 Suggested solutions

Question 11

Marker comments

$$\begin{aligned} a) \quad i) \quad 2z + 3\bar{w} &= 2(5+5i) + 3(2-i) \\ &= 10 + 10i + 6 - 3i \\ &= 16 + 7i \end{aligned}$$

$$\begin{aligned} ii) \quad z(w-i) &= (5+5i)(1+i) \\ &= 5 + 5i + 5i - 5 \\ &= 10i \end{aligned}$$

$$\begin{aligned} iii) \quad \frac{z}{w} &= \frac{5+5i}{2+i} = \frac{(5+5i)(2-i)}{(2+i)(2-i)} \\ &= \frac{10 - 5i + 10i + 5}{4+1} \\ &= \frac{15 + 5i}{5} \\ &= 3 + i \end{aligned}$$

$$\begin{aligned} iv) \quad \arg z - \arg \bar{z} &= \tan^{-1}\left(\frac{5}{5}\right) + \tan^{-1}\left(\frac{5}{5}\right) \\ (-\arg z) &= \frac{\pi}{4} + \frac{\pi}{4} \\ &= \frac{\pi}{2} \end{aligned}$$

$$\begin{aligned} b) \quad \frac{(1+\sqrt{3}i)^6}{(1-i)^4} &= \frac{(2 \operatorname{cis} \frac{\pi}{3})^6}{(\sqrt{2} \operatorname{cis} (-\frac{\pi}{4}))^4} \\ &= \frac{2^6 \operatorname{cis} 2\pi}{2^2 \operatorname{cis} (-\pi)} \\ &= 2^4 \operatorname{cis} 3\pi \\ &= 16 \times (-1) \\ &= -16. \end{aligned}$$

Question 11 c) d)

Marker comments

c) Let $z^3 = 1$ and ω is a root $\omega \in \mathbb{C}$.

$$\therefore z^3 - 1 = (z - 1)(z^2 + z + 1)$$

$$\Rightarrow \omega^2 + \omega + 1 = 0 \quad (1) \text{ since } \omega \text{ is a root}$$

$$\Rightarrow \omega^3 = 1 \quad (2)$$

$$(1 - 3\omega + \omega^2)(1 + \omega - 8\omega^2)$$

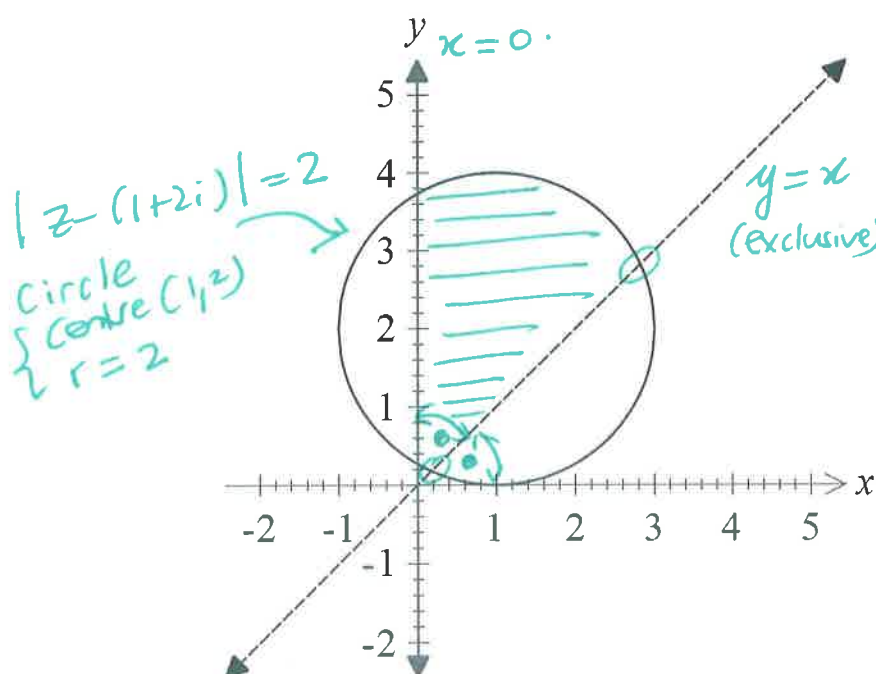
$$= (1 - 3\omega + (-\omega - 1))(\omega^2 - 8\omega^2) \text{ from (1)}$$

$$= (-4\omega)(-9\omega^2)$$

$$= (-4)(-9)\omega^3$$

$$= 36$$

d)

✓ $\rightarrow$ circle✓ $\rightarrow y = x$ (excluding boundary)✓ $\rightarrow$ shaded region.

Question 12

Marker comments

$$a) \quad z^6 = -64$$

$$= 64 \cos \pi$$

$$\therefore z = 64^{1/6} (\cos \pi)^{1/6}$$

$$= (2^6)^{1/6} \cos (\pi + 2k\pi)^{1/6}$$

$$= 2 \cos \left(\frac{\pi + 2k\pi}{6} \right) \quad k \in \mathbb{Z}$$

$$z_1 = 2 \cos \pi/6 \quad (k=0)$$

$$z_2 = 2 \cos \left(\frac{3\pi}{6} \right) = 2 \cos \frac{\pi}{2} \quad (k=1)$$

$$z_3 = 2 \cos \left(-\frac{\pi}{6} \right) \quad (k=-1) \rightarrow \text{could use}$$

$$k=5 \Rightarrow z = 2 \cos \left(\frac{11\pi}{6} \right)$$

$$z_4 = 2 \cos \left(\frac{5\pi}{6} \right) \quad (k=2)$$

$$z_5 = 2 \cos \left(-\frac{\pi}{2} \right) \quad (k=-2) \rightarrow \text{could use}$$

$$k=4, \Rightarrow z = 2 \cos \frac{3\pi}{2}$$

$$z_6 = 2 \cos \left(\frac{\pi}{6} \right) \quad (k=3) \rightarrow \text{could use}$$

$$k=-3, \Rightarrow z = 2 \cos \left(-\frac{5\pi}{6} \right)$$

[NB: all roots are equally spaced on the circle of radius 2, centre (0,0) by $\frac{\pi}{3}$ at the centre].

$$b) \quad \int_0^{\pi/2} \sin^3 x \, dx = \int_0^{\pi/2} \sin^2 x \cdot \sin x \, dx$$

$$= \int_0^{\pi/2} (1 - \cos^2 x) \sin x \, dx$$

$$= \int_0^{\pi/2} \sin x - \sin x (\cos x)^2 \, dx$$

$$= \left[-\cos x + \frac{\cos^3 x}{3} \right]_0^{\pi/2}$$

$$= \left[-\cos \frac{\pi}{2} + \frac{(\cos \frac{\pi}{2})^3}{3} + \cos 0 - \frac{\cos 0}{3} \right]$$

$$= 0 + 0 + 1 - \frac{1}{3} = \frac{2}{3}$$

Question 12 ctd.

Marker comments

$$c) i) x^2 + x + 1 \equiv Ax(x+3) + B(x+3) + Cx^2$$

$$\text{let } x=0, \Rightarrow 1 = 3B \\ \therefore B = \frac{1}{3}$$

$$\text{let } x=-3, \Rightarrow 9-3+1=9C \\ \therefore C = \frac{7}{9}$$

$$\text{let } x=1, \Rightarrow 3 = A \times 1 \times 4 + B \times 4 + C$$

$$\underline{\text{we}} \quad 3 = 4A + \frac{4}{3} + \frac{7}{9}$$

$$\Rightarrow A = \frac{2}{9}$$

$$\therefore (A, B, C) = \left(\frac{2}{9}, \frac{1}{3}, \frac{7}{9}\right)$$

$$ii) \int \frac{x^2 + x + 1}{x^3 + 3x^2} dx = \int \frac{\frac{2}{9}}{x} + \frac{\frac{1}{3}}{x^2} + \frac{\frac{7}{9}}{x+3} dx$$

$$= \frac{2}{9} \ln|x| - \frac{1}{3x} + \frac{7}{9} \ln|x+3| + C$$

from (i)

$$d) i) x^2 + x + 1 = x^2 + x + \left(\frac{1}{2}\right)^2 + 1 - \left(\frac{1}{2}\right)^2 \\ = \left(x + \frac{1}{2}\right)^2 + \frac{3}{4}$$

$$ii) \int \frac{1}{x^2 + x + 1} dx = \int \frac{1}{\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}} dx \\ = \frac{2}{\sqrt{3}} \tan^{-1} \frac{2\left(x + \frac{1}{2}\right)}{\sqrt{3}} + C$$

$$iii) t = \tan \frac{x}{2}$$

$$\frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$$

$$= \frac{1}{2} (\tan^2 \frac{x}{2} + 1)$$

$$\therefore \frac{dt}{dx} = \frac{1}{2} (t^2 + 1)$$

2015 MAXX Half-Yearly Task 3 Suggested solutions

Question 12(d) ctd

$$\therefore dx = \frac{2}{1+t^2} dt$$

hence $\int \frac{1}{2+\sin x} dx$

$$= \int \frac{1}{2 + \frac{2t}{1+t^2}} \cdot \frac{2}{1+t^2} dt$$

$$= \int \frac{1}{t^2 + t + 1} dt$$

$$= \frac{2}{\sqrt{3}} \tan^{-1} \left[\frac{2(t + \frac{1}{2})}{\sqrt{3}} \right] + C \quad \text{from (ii)}$$

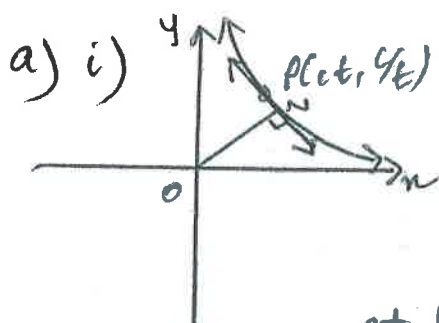
$$= \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2 \tan \frac{x}{2} + 1}{\sqrt{3}} \right) + C$$

Marker comments



Question 13

Marker comments



$$xy = c^2$$

$$\therefore y(1) + xy' = 0$$

$$\therefore \frac{dy}{dx} = -\frac{y}{x}$$

$$\text{at } P, \frac{dy}{dx} = \frac{c/t}{ct} = -\frac{1}{t^2}$$

$\therefore$ Equation of tangent is
 $y - \frac{c}{t} = -\frac{1}{t^2}(x - ct)$

$$\therefore yt^2 - ct = -x + ct$$

$$\therefore x + yt^2 = 2ct$$

ii) Equation of ON is $y = t^2x$ ①
 ($m_1 m_2 = -1$ at N $\therefore$ slope of ON = t^2).

Solving: $\begin{cases} y = t^2x & \text{①} \\ x + yt^2 = 2ct & \text{②} \end{cases}$

Sub ① into ② $\Rightarrow x + (t^2x)t^2 = 2ct$

$$\Rightarrow x(1 + t^4) = 2ct$$

$$\therefore x = \frac{2ct}{1+t^4} \quad \text{③}$$

Sub ③ into ① $\therefore y = t^2 \left(\frac{2ct}{1+t^4} \right)$

$$y = \frac{2ct^3}{1+t^4} \quad \text{④}$$

$$\therefore N = \left(\frac{2ct}{1+t^4}, \frac{2ct^3}{1+t^4} \right)$$

Question13(a) ctd.

Marker comments

$$\text{iii) From } \textcircled{3} \text{ and } \textcircled{4} \quad \begin{cases} x = \frac{2ct}{1+t^4} \rightarrow \textcircled{3} \\ y = \frac{2ct^3}{1+t^4} \rightarrow \textcircled{4} \end{cases}$$

and From $\textcircled{1}$ $\frac{y}{x} = t^2$

ie $t^4 = \frac{y^2}{x^2} \rightarrow \textcircled{5}$

in $\textcircled{3}$ $x = \frac{2c(1+t^4)}{2c}$
 $= \frac{2c(1 + \frac{y^2}{x^2})}{2c}$ from $\textcircled{5}$

$t = \frac{x^2 + y^2}{2cx}$

hence in $\textcircled{4}$

$y(1+t^4) = 2ct^3$

ie $y[1 + \frac{y^2}{x^2}] = 2c[\frac{x^2 + y^2}{2cx}]^3$

$\frac{y(x^2 + y^2)}{x^2} = \frac{2c}{8c^3x^3} (x^2 + y^2)^3$

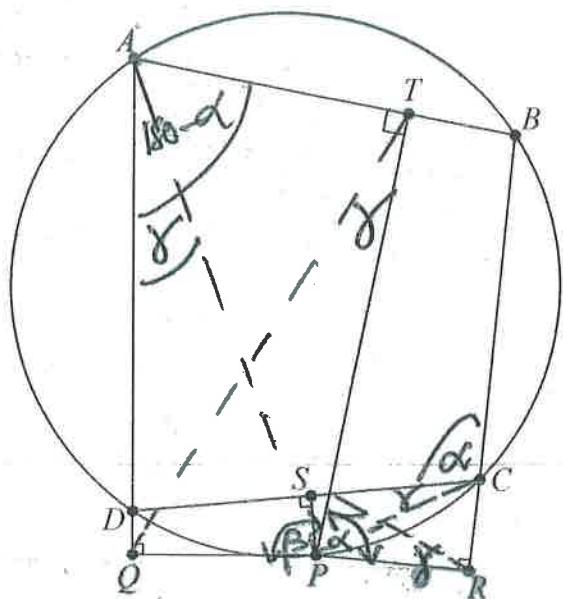
ie $\frac{y}{x^2} = \frac{1}{4c^2x^3} (x^2 + y^2)^2$

ie $4c^2xy = (x^2 + y^2)^2$

Question 13(6)

Marker comments

i)



In quad $SPRC$, $\angle DSP = \angle CRP = 90^\circ$
(exterior angle of a quad)

As the ext $\angle$ at S is equal to the interior opposite angle at R , then $SPRC$ is a cyclic quad.

Also $\angle ATP = \angle PQA = 90^\circ$

Since $\angle ATP + \angle PQA = 90 + 90$
 $= 180^\circ$

then $AQPT$ is a cyclic quad
(opp. $\angle$'s are supplementary).

ii) Let $\angle SPR = \alpha$, $\angle QPT = \beta$

Join SR , PC , TQ , PA

$\angle BCS = \alpha$ (ext $\angle$ of cyclic quad $SPRC$)

$\therefore \angle DAB = 180 - \alpha$ (opp. supp. $\angle$'s in cyclic quad $ABCD$)

$\therefore \angle TPQ = \alpha$ (opp. supp. $\angle$'s, $AQPT$ cyclic)
 $= \beta$

$\therefore \alpha = \beta \quad \therefore \angle SPR = \angle QPT$

Question 13(b) ctd.

Marker comments

Let $\angle PRS = x$ $\therefore \angle SCP = x$ (Δ 's in same segment $SPRC$ cyclic)also $\angle DAP = x$ (Δ 's in same segment $ACPD$ cyclic) $= \angle QTP$ (Δ 's in same segment $AQPT$ cyclic) $\therefore \angle PRS = \angle QTP$ iii) RTP: $\triangle SPR \parallel \triangle QPT$ 1. $\angle SPR = \angle QPT = \alpha$ (see ii)2. $\angle PRS = \angle PTQ = x$ (see ii) $\therefore \triangle SPR \parallel \triangle QPT$ (equiangular)iv) (a) Since corresponding sides in $\parallel \Delta$'s are in proportion then

$$\frac{PS}{PQ} = \frac{PR}{PT}$$

$$\therefore (PS)(PT) = (PR)(PQ)$$

$$(b) \text{ also: } \frac{PS}{PQ} = \frac{PR}{PT} = \frac{SR}{QT}$$

(corresponding sides proportional)

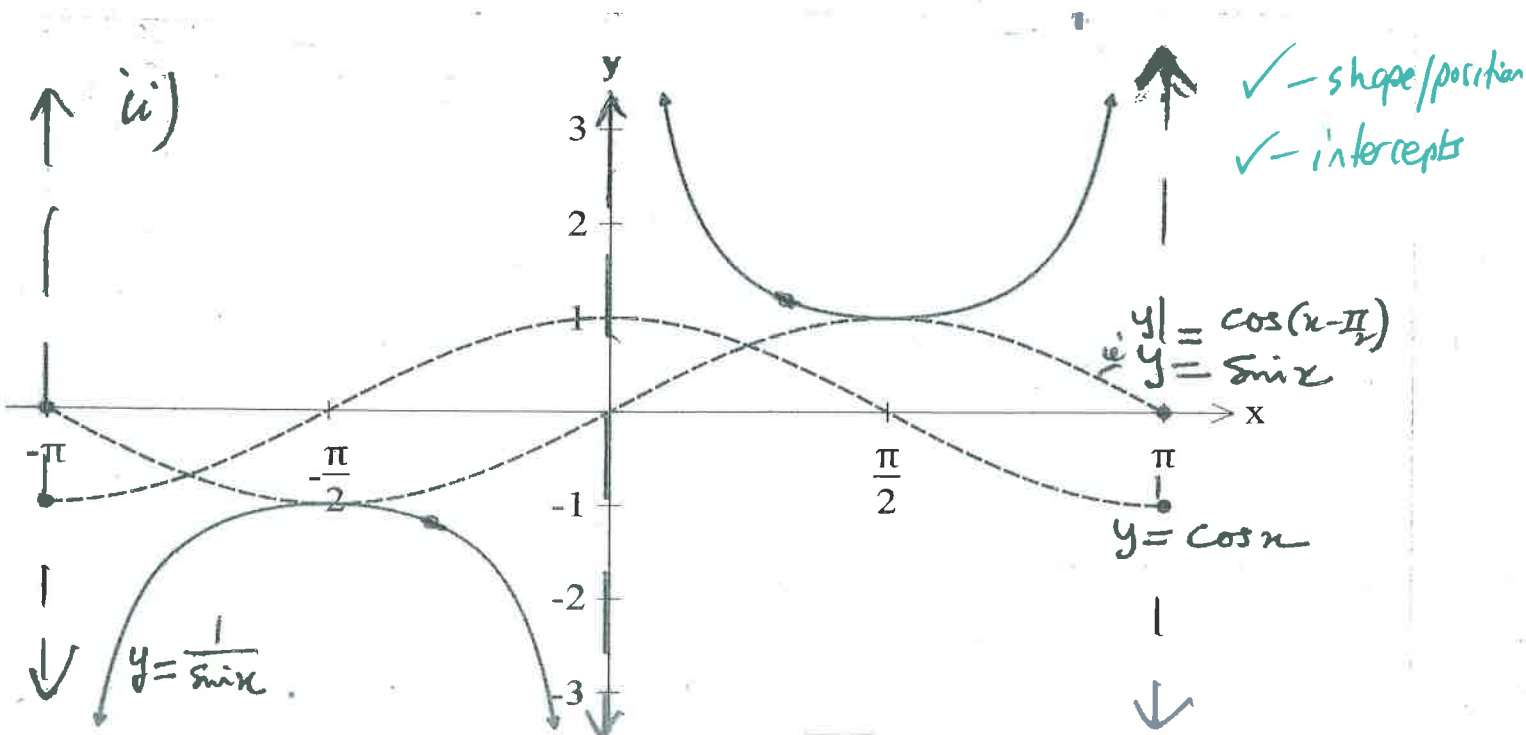
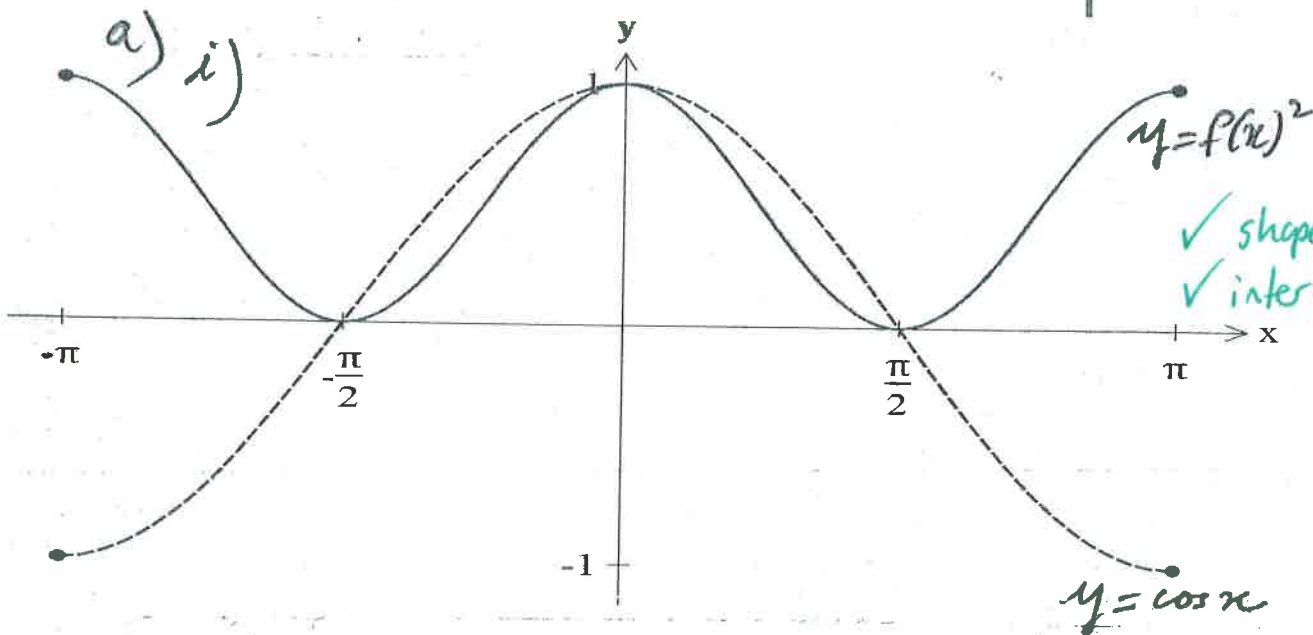
$$\therefore \frac{PS}{PQ} \cdot \frac{PR}{PT} = \frac{SR}{QT} \cdot \frac{SR}{QT}$$

$$\therefore \frac{PS}{PQ} \cdot \frac{PR}{PT} = \left(\frac{SR}{QT}\right)^2$$

* All graphs must be within the specified domain *

Question 14

Marker comments

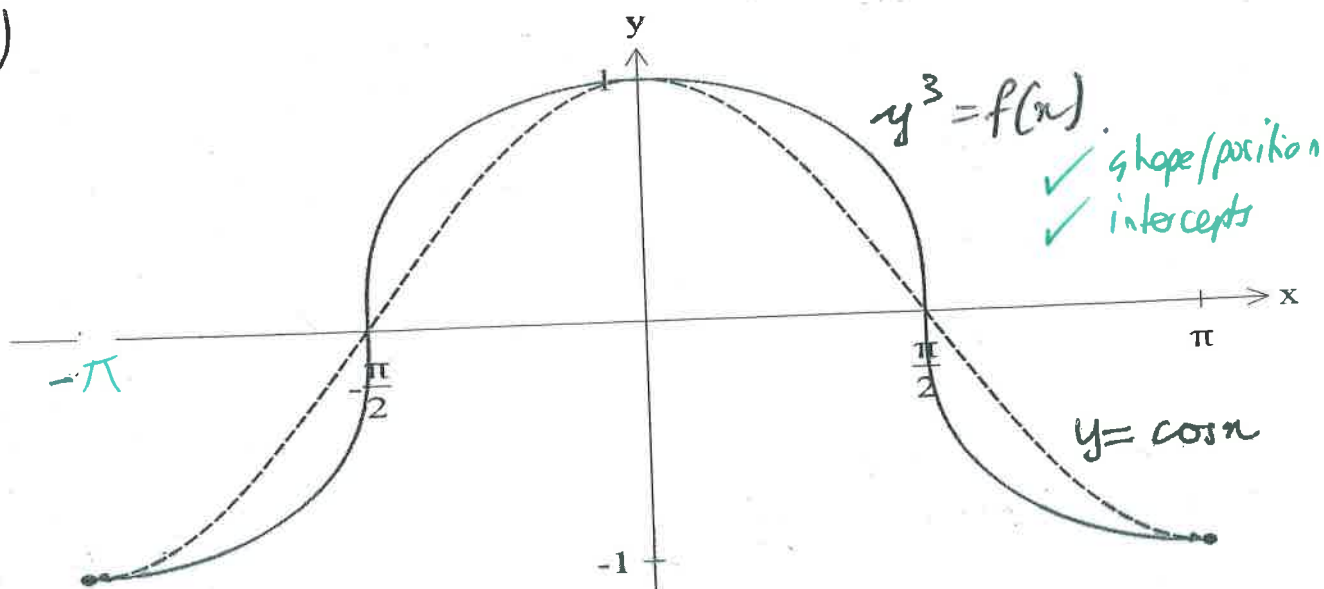


2015 MAXX Half-Yearly Task 3 Suggested solutions

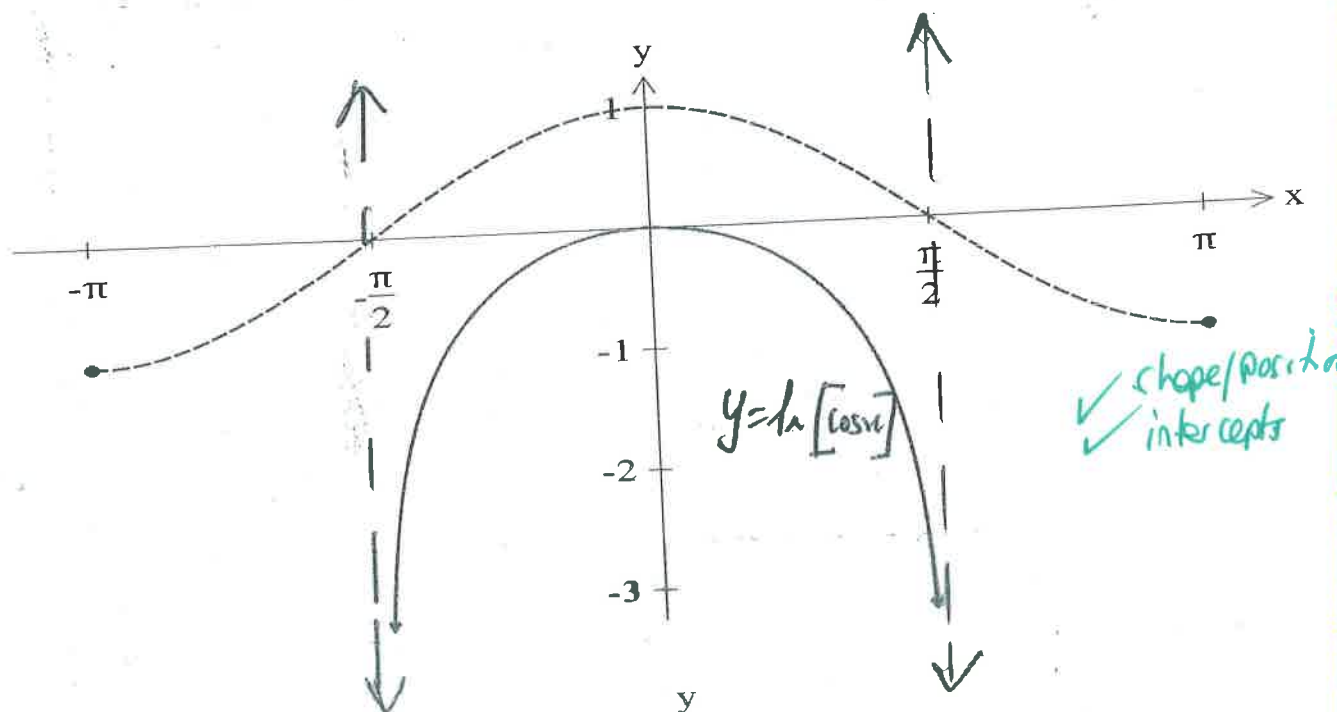
Question 14(a) ctd.

Marker comments

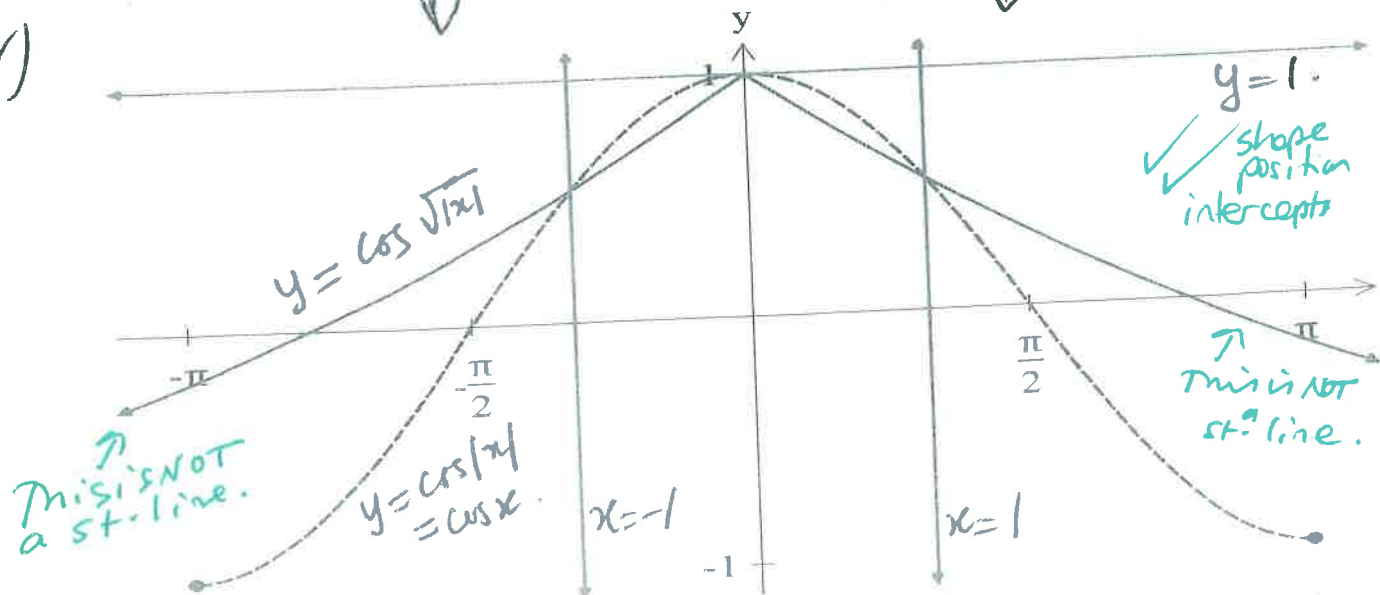
iii)



iv)

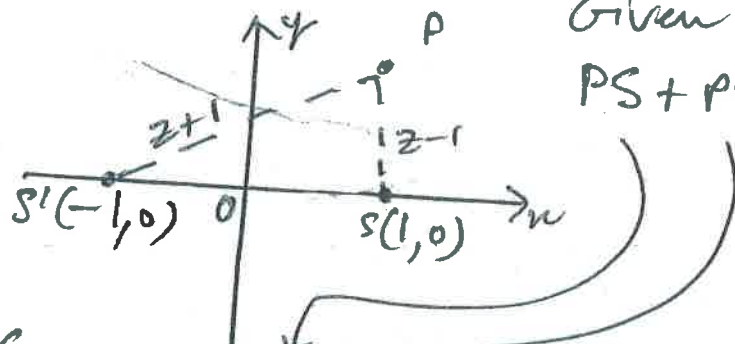


v)



Question 14(b)

b)



Given definition
 $PS + PS' = 2a$

Marker comments

Given $|z-1| + |z+1| = 6$

$\therefore 2a = 6$

$\Rightarrow a = 3$

we know $ae = 1$

$\therefore e = \frac{1}{a}$

$e = \frac{1}{3}$

Also

$b^2 = a^2(1 - e^2)$ [$0 < e < 1$]

$\therefore b^2 = 9(1 - \frac{1}{9})$

$b^2 = 9 - 1$

$b^2 = 8$

hence $\frac{x^2}{9} + \frac{y^2}{8} = 1$

[If the given definition was not used
a maximum of 2 marks is awarded]

[Alternate: $|(x-1) + yi| + |(x+1) + yi| = 6$

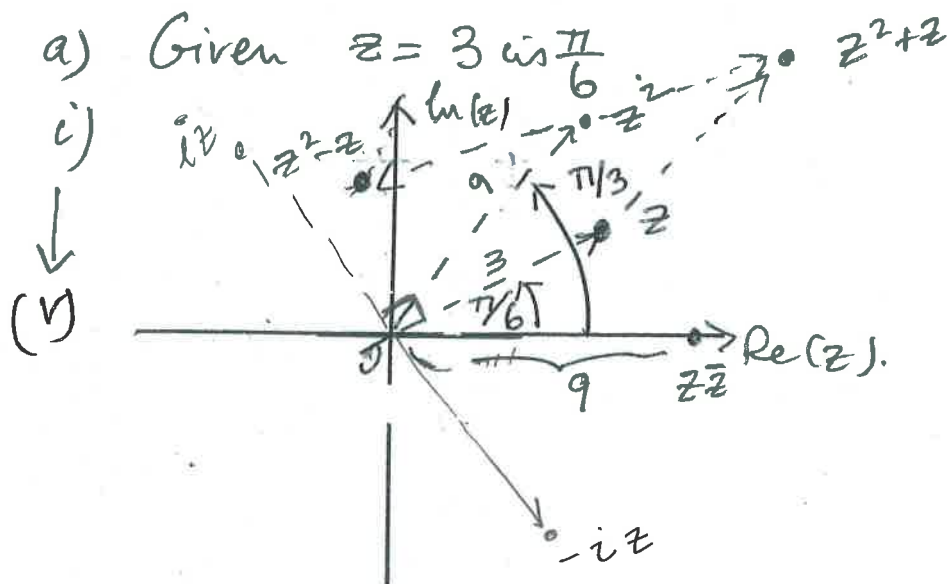
$\Rightarrow \sqrt{(x-1)^2 + y^2} + \sqrt{(x+1)^2 + y^2} = 6$ (not 36)

$\therefore (\sqrt{(x-1)^2 + y^2} - 6)^2 = (-\sqrt{(x+1)^2 + y^2})^2$

$\Rightarrow (9 - x) = 3\sqrt{(x-1)^2 + y^2}$

$\Rightarrow 9x^2 - 18x + 9 + 9y^2 = 0$ etc...

Question 15



$$-iz = -iz \quad [25 \div 4 = 6r1]$$

$$iz = 3 \cos \left(\frac{\pi}{6} + \frac{\pi}{2} \right) = 3 \cos \left(\frac{2\pi}{3} \right)$$

$$\therefore -iz = 3 \cos \left(-\frac{\pi}{3} \right) = \left(3 \cos \left(-\frac{\pi}{3} \right) \right)$$

$$z\bar{z} = |z|^2 = 3^2 = 9 \quad [\text{or } 9 \cos 0].$$

iv) $z^2 = 9 \cos \frac{\pi}{3}$

$\therefore z^2 + z$ is drawn by vector addition

$z^2 - z$ see diagram

b) $|z-1| = \text{Re}(z)$

$$\Rightarrow |(x-1) + yi| = x$$

$$\underline{\text{we}} \quad (x-1)^2 + y^2 = x^2$$

$$\underline{\text{we}} \quad y^2 = 2x - 1 \quad (\text{on expansion})$$

$$y^2 = 2\left(x - \frac{1}{2}\right) \quad \text{or} \quad y^2 = 4\left(\frac{1}{2}\right)\left(x - \frac{1}{2}\right)$$

This is a parabola $V\left(-\frac{1}{2}, 0\right)$ and $|a| = \frac{1}{2}$.

Marker comments

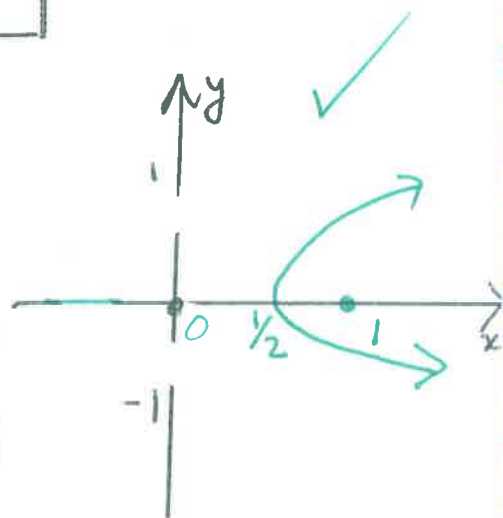
✓ z

✓ $-i^{25}z$

✓ $z\bar{z}$

✓ $z^2 + z$

✓ $z^2 - z$



Question 15(c)

$$c) H: \frac{y^2}{4} - \frac{x^2}{12} = 1$$

$$2) i) \text{ let } a^2 = 4 \\ b^2 = 12$$

$$\therefore b^2 = a^2(e^2 - 1) \quad \text{hyperbola}$$

$$12 = 4(e^2 - 1)$$

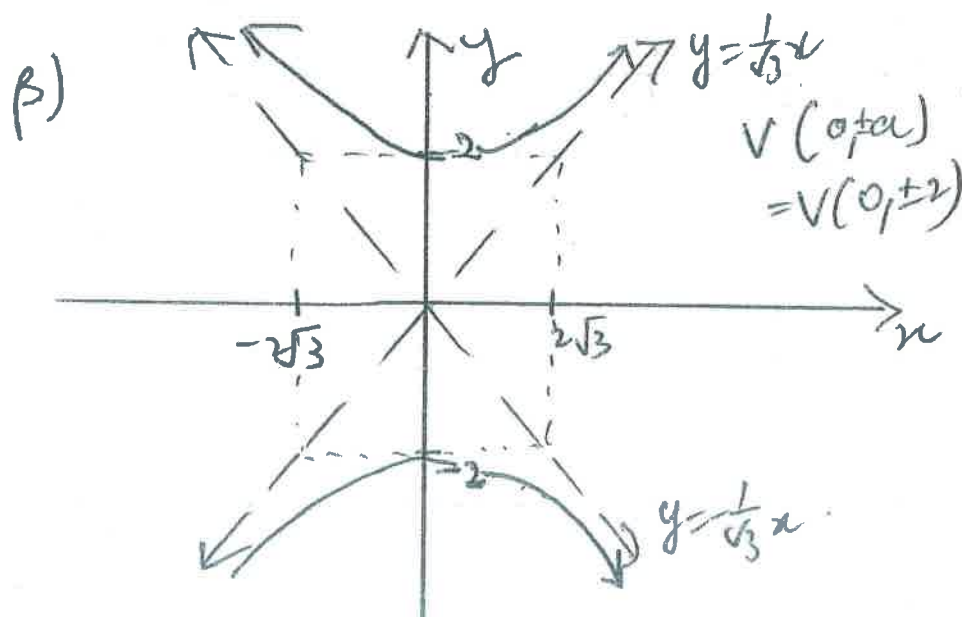
$$e^2 = 4$$

$$\therefore e = 2$$

$$ii) S(0, \pm ae) = S(0, \pm 4)$$

$$iii) y = \pm \frac{a}{e} = \pm 1$$

$$iv) y = \pm \frac{a}{b}x = \pm \frac{1}{\sqrt{3}}x$$



Marker comments



✓ shape/position

✓ labelled intercepts/asymptotes.

2015 MAXX Half-Yearly Task 3 Suggested solutions

Question^{15(d)}

Marker comments

$$\begin{aligned} d) \quad & \int_1^2 \frac{1}{x^2} e^{\frac{1}{x}} dx \\ &= \int_1^2 -\frac{d}{dx} \left(\frac{1}{x} \right) e^{\frac{1}{x}} dx \\ &= -e^{\frac{1}{x}} \Big|_1^2 \\ &= -(e^{\frac{1}{2}} - e^1) \\ &= -(\sqrt{e} - e) \\ &= e - \sqrt{e} \end{aligned}$$



Question 16.

$$a) P = [a \cos \theta, b \sin \theta]$$

$$\begin{aligned} i) Q &= [a \cos(\theta + \frac{\pi}{2}), b \sin(\theta + \frac{\pi}{2})] \\ &= [a(\cos \theta \cos \frac{\pi}{2} - \sin \theta \sin \frac{\pi}{2}), \\ &\quad b(\sin \theta \cos \frac{\pi}{2} + \sin \frac{\pi}{2} \cos \theta)] \\ &= [-a \sin \theta, b \cos \theta] \end{aligned}$$

$$\begin{aligned} ii) LHS &= OP^2 + OQ^2 \\ &= a^2 \cos^2 \theta + a^2 \sin^2 \theta + b^2 \sin^2 \theta + b^2 \cos^2 \theta \\ &= a^2 (\cos^2 \theta + \sin^2 \theta) + b^2 (\sin^2 \theta + \cos^2 \theta) \\ &= a^2 \cdot 1 + b^2 \cdot 1 \\ &= a^2 + b^2 \\ &= RHS \end{aligned}$$

$$iii) M = \left[\frac{-a \sin \theta + a \cos \theta}{2}, \frac{b \cos \theta + b \sin \theta}{2} \right]$$

$$ie M = \left[\frac{a}{2} (\cos \theta - \sin \theta), \frac{b}{2} (\cos \theta + \sin \theta) \right]$$

$$\begin{aligned} LHS &= \frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{a^2}{4a^2} (\cos^2 \theta - 2 \sin \theta \cos \theta + \sin^2 \theta) \\ &\quad + \frac{b^2}{4b^2} (\cos^2 \theta + 2 \sin \theta \cos \theta + \sin^2 \theta) \\ &= \frac{1}{4} [1 + 1 + 0] \\ &= \frac{2}{4} \\ &= \frac{1}{2} = RHS \end{aligned}$$

Marker comments

No marks for substituting Q into E.

Only 1 mark for substituting $(\theta + \frac{\pi}{2})$ instead of θ and getting straight to solution.

(OR)

$$Q = (a \cos(\frac{\pi}{2} - \theta), b \sin(\frac{\pi}{2} - \theta))$$

$$= (a \sin(-\theta), b \cos(-\theta))$$

$$= (-a \sin \theta, b \cos \theta).$$

needed to be substituted into equation

Question 11b a) ctd.

Marker comments

$$\text{iv) } T_P: \frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1 \quad (1)$$

$$\text{v) } T_Q: -\frac{x}{a} \sin \theta + \frac{y}{b} \cos \theta = 1 \quad (2)$$

Solving (1) & (2) for x, y

$$\text{in (1) } (x) - \frac{\sin \theta}{a} \Rightarrow \frac{-x}{a} \sin \theta \cos \theta - \frac{y \sin^2 \theta}{ab} = -\frac{\sin \theta}{a} \quad (3)$$

$$\text{in (2) } (x) \frac{\cos \theta}{a} \Rightarrow \frac{-x \sin \theta \cos \theta}{a} + \frac{y \cos^2 \theta}{ab} = \frac{\cos \theta}{a} \quad (4)$$

$$(4) - (3) \quad \frac{y}{ab} (\underbrace{\cos^2 \theta + \sin^2 \theta}_{=1}) = \frac{1}{a} (\cos \theta + \sin \theta)$$

$$\therefore y = b (\cos \theta + \sin \theta) \quad (5)$$

Sub (5) into (1)

$$\frac{x}{a} \cos \theta + \frac{b (\cos \theta + \sin \theta) \sin \theta}{b} = 1$$

$$\frac{x \cos \theta}{a} + \cos \theta \sin \theta + \sin^2 \theta = 1$$

$$x \cos \theta = a (1 - \sin^2 \theta - \cos \theta \sin \theta)$$

$$x \cos \theta = a (\cos^2 \theta - \cos \theta \sin \theta)$$

$$x \cos \theta = a (\cos \theta) (\cos \theta - \sin \theta)$$

$$\therefore x = a (\cos \theta - \sin \theta)$$

$\therefore$ The point of intersection T is

$$[a (\cos \theta - \sin \theta), b (\cos \theta + \sin \theta)]$$

Question 16(a) ctd

Marker comments

$$vi) LHS = \frac{x^2}{a^2} + \frac{y^2}{b^2}$$

$$= \frac{a^2}{a^2} (\cos\theta - \sin\theta)^2 + \frac{b^2}{b^2} (\sin\theta + \cos\theta)^2$$

$$= \cos^2\theta - 2\sin\theta\cos\theta + \sin^2\theta + \sin^2\theta + 2\sin\theta\cos\theta + \cos^2\theta$$

$$= 2(\cos^2\theta + \sin^2\theta)$$

$$= 2 \times 1$$

$$= 2$$

$$= RHS$$

$$b) \text{ Given } 1+2+3+\dots+k = \frac{k(k+1)}{2} \quad (1)$$

$$\text{Let } k = 15n \text{ from (1)}$$

$$\therefore \sum_{k=1}^{15n} k = \frac{15n(15n+1)}{2} \quad (2)$$

[This is the sum of all integers from $1 \rightarrow 15n$ inclusive].

We must now eliminate from this sum the sum of integers divisible by 3 or 5.

$$\text{By 3: } 3+6+9+\dots+15n \quad \leftarrow \text{since } 15n \text{ is } \div \text{ by } 3$$

Number of terms given by $T_k = a + (k-1)d$

$$15n = 3 + (k-1)3$$

$$\therefore k = 5n \text{ (terms)}$$

2015 MAXX Half-Yearly Task 3 Suggested solutions

Question 16/65 ctd.

Marker comments

$$\begin{aligned}
 \therefore 3+6+9+\dots+15n \\
 &= 3(1+2+3+\dots+5n) \\
 &= 3 \sum_{k=1}^{5n} k \\
 &= 3 \left(\frac{5n(5n+1)}{2} \right) \\
 &= \frac{15n(5n+1)}{2} \quad \text{--- (3)}
 \end{aligned}$$

Similarly the sum of the integers $\div$ by 5 are:

$$\begin{aligned}
 5+10+\dots+15n \\
 &= 5(1+2+3+\dots+3n) \\
 &= 5 \times \sum_{k=1}^{3n} k \quad \nwarrow (3n \text{ terms}) \\
 &= 5 \left(\frac{3n(3n+1)}{2} \right) \\
 &= \frac{15n(3n+1)}{2} \quad \text{--- (4)}
 \end{aligned}$$

However, by subtracting (3) & (4) from (2) results in removing that are divisible by 15 twice so we need to add one set back in.

$$\begin{aligned}
 \text{So, } 15+30+45+\dots+15n \\
 &= 15(1+2+3+\dots+n) \\
 &= 15 \sum_{k=1}^n k
 \end{aligned}$$

Question ...16(b)

Marker comments

$$= \frac{15n(n+1)}{2} \quad \text{--- (5)}$$

Hence, the sum of the integers which are not divisible by 3 or 5 is given

$$\text{by } \left[\frac{15n(15n+1)}{2} \right] - \left[\frac{15n(3n+1)}{2} \right] - \left[\frac{15n(5n+1)}{2} \right] + \left[\frac{15n(n+1)}{2} \right]$$

$$= \frac{15n}{2} [15n+1 - 3n-1 - 5n-1 + n+1]$$

$$= \frac{15n}{2} (8n)$$

$$= 60n^2$$

$$c) \ i) \ \sin(x-\theta) = k \sin(x+\theta)$$

$$\sin x \cos \theta - \sin \theta \cos x = k(\sin x \cos \theta + \sin \theta \cos x)$$

$$\sin x \cos \theta - k \sin x \cos \theta = k \sin \theta \cos x + \sin \theta \cos x$$

$$\sin x \cos \theta (1-k) = \sin \theta \cos x (k+1)$$

$$\Rightarrow \frac{\sin x}{\cos x} = \frac{\sin \theta}{\cos \theta} \frac{(k+1)}{(1-k)}$$

$$\text{ie } \tan x = \tan \theta \frac{(k+1)}{-(k-1)}$$

$$\text{ie } \tan x = -\frac{(k+1)}{k-1} \tan \theta$$

2015 MAXX Half-Yearly Task 3 Suggested solutions

Question 16(c) *std.*

Marker comments

ii) $k = \frac{1}{2}, \theta = 210^\circ$

$$\therefore \tan x = \frac{-(\frac{3}{2}) \tan 210^\circ}{(-\frac{1}{2})}$$

Substitution needed.

$$\tan x = 3 \times \tan 30^\circ$$

$$\begin{aligned} \tan x &= \frac{3}{\sqrt{3}} \\ &= \frac{3}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} \end{aligned}$$

$$\tan x = \sqrt{3}$$

$$\Rightarrow x = 60^\circ, 240^\circ$$

(Radians not accepted)